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Monitoring collegiate soccer players during a congested match schedule: Heart rate variability versus subjective wellness measures

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ABSTRACT

The aims of this study were a) to examine within-group changes of wellness and heart rate variability measures and b) to compare their sensitivity to a congested match schedule in collegiate soccer players (n = 8). Wellness (Hooper index and its subsets) and heart rate variability (Ln rMSSD, SDNN) measures were assessed after selected low-load (training sessions) and high-load (a congested match schedule) phases. Session rating of perceived exertion (sRPE) was computed for training and match sessions. A *very likely* large difference in accumulated sRPE was observed between low-load and high-load phases (+148.4%, 90% confidence interval CI [87.3; 229.5%]); effect size, ES, 2.16 [1.49; 2.82]. While the Hooper index showed an *almost certainly* moderate increase (+49.8%, [33.9; 67.5%]), ES, 1.05 [0.76; 1.34], heart rate variability measures (i.e., Ln rMSSD and SDNN) only changed with a *possible* trivial effect (range −2.1; 8.2%, [−7.1; 16.7%]), ES, −0.15; 0.15 [−0.50; 0.44]. The Hooper index showed a moderately higher sensitivity than Ln rMSSD to a congested match schedule (34.7%, [26.9; 41.6%], ES, 0.81 [0.60; 1.03]). Relationships between changes in the Hooper index and some of its subsets (ΔHooper index, Δsleep, and Δfatigue), with changes in mean sRPE (ΔsRPE) were very large (range $r = 0.72$; 0.89). However, small associations were observed between changes in heart rate variability (ΔLn rMSSD, and ΔSDNN) and ΔsRPE (range $r = -0.21$; 0.10). This study suggests the use of subjective wellness indices, instead of heart rate variability measures, to monitor collegiate soccer players during congested match schedules.

1. Introduction

Monitoring soccer players' physical fatigue is of paramount importance for practitioners aiming to optimize training loads [1,2] and to prevent possible fatigue-induced injuries [3,4]. A myriad of measures, including blood and salivary hormones [5,6], short-duration maximal performances [7,8], submaximal heart rate measures [9], resting heart rate variability indices [10,11], and self-reported measures [12,13], have been examined by researchers to monitor fatigue in soccer players.

While collecting laboratory-based biomarkers is of special interest among the scientific community, sports scientists working in club settings are always looking to find more time-efficient and easy-to-use measures to monitor athletes' training status (fitness and/or fatigue) [14]. The aim of collecting such measures is to avoid situations such as overtraining, to mitigate injury risks, and to optimize human performance [1,4]. Therefore, the interest in more practical measures of fatigue monitoring has recently been growing [2,12,15,16], with an aim to help practitioners to solve problems arising from real-world scenarios (i.e., club settings).

Subjective self-reported measures are among the top variables commonly used in practice as a useful monitoring tool [1]. More specifically, the Hooper index and/or its subsets (i.e., sleep quality, stress, fatigue, and muscle soreness) [17] have recently been shown as a promising measure of monitoring in soccer players [8,12,13,18–20]. Indeed, self-reported measures are not only pertinent to high-load versus low-load training phases [13,18,20,21] but are also associated with training load fluctuations [8,20] in soccer players.

Despite laboratory-based objective measures (e.g., blood and salivary hormones), field-based alternatives, like heart rate variability obtained using smart-phone applications, have recently received special attention among practitioners for monitoring soccer players' training status [11,15,16,22]. The natural logarithm of the square root of the

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mean of the sum of the squares of differences between adjacent normal R-R intervals (Ln rMSSD) has been recommended as a strong heart rate variability index for monitoring athletes' training status [2], and its sensitivity to training adaptations [22] and training-induced fatigue [8,19] has been evidenced among soccer players.

Recent studies have shown that Hooper index and heart rate variability show promising practical applications for monitoring soccer players' training status. Hooper index has been shown to be influenced by match-induced loads in academy soccer players [12]. Moalla et al. [20] observed that all Hooper index subsets, including perceived sleep, stress, fatigue and muscle soreness are moderately related to daily training load in professional soccer players. Fessi et al. [18] also observed that all subsets of the Hooper index show higher values (detrimental change) in high-load versus low-load phases (pre-season vs. in-season). The use of Ln rMSSD not only has practical applications for monitoring training-induced adaptations [11] but is also a practical measure to detect fatigue [10,16] in soccer players. Thorpe et al. [8] showed that fluctuations in Ln rMSSD are significantly related to fluctuations in external load (i.e., total high-intensity running distance) in elite soccer players.

The importance of monitoring players' training status and their sensitivity to high-load periods is even greater during congested fixtures [21,23]. Usually, congested fixtures are characterized by a specific high-load period with matches occurring every three days or less. Higher internal or external training loads have been observed when comparing regular with congested fixtures in some studies examining session-ratings of perceived exertion (sRPE) [13] or performance variables such as frequency of accelerations [24]. However, while changes in both wellness and heart rate variability measures to changes in training load have been usually investigated in preseason preparations [11,16], high-load versus low-load periods [19,22], and in-season phases [8], their sensitivity to congested match schedules has not been compared yet. Determining the most sensitive measure of congested match schedules helps practitioners select the best option for their monitoring purposes. Therefore, the aims of this study were a) to examine within-group changes in wellness and heart rate variability measures and b) to compare their sensitivity to a congested match schedule in collegiate soccer players.

2. Materials and methods

2.1. Experimental approach

Soccer players' wellness measures were collected after four consecutive days of low-load (training weak) and high-load (congested match schedule) phases, at the same time (i.e., ~10 a.m.). Heart rate variability was also collected by players using a smart-phone application based on their individual waking times (i.e., 7:00 to 9;00 a.m.) on the same testing days. Wellness measures were also reported by players at the same time they provided heart rate variability data. The results of both heart rate variability and Hooper index were sent to the investigator immediately after their collection with a smartphone application (WhatsApp) by all individual players. Two weeks before commencing the experiment, all players were familiarized with measuring heart rate variability using the smart-phone application. During all training and match sessions, rating of perceived exertion and total training times were collected to estimate internal training loads. The sensitivity of heart rate variability and Hooper index were examined by analyzing their standardized changes and the standardized differences between their changes, as well as the relationship between their changes (i.e., so-called Δ vs. Δ relationship).

2.2. Subjects

Data from eight collegiate soccer payers (21.1 ± 3.4 yrs., 72.4 ± 4.2 kg, 177.2 ± 5.4 cm, training experience 5.2 ± 2.1 yrs) were collected in this study. All participants were members of semi-professional or professional soccer teams and had regular weekly training sessions with their clubs for the last six months prior to collegiate training. However, for the one month immediately prior to the National Collegiate Tournament, they only participated in three to four weekly training sessions, with the same collegiate team preparing for the tournament. The players ranked first in the National Iranian Collegiate Tournament at the time of the study. All players were informed of the experimental risks, and all signed an informed consent document before the investigation began. The protocol was approved by the local research ethics committee, and the study conformed with the Declaration of Helsinki.

2.3. Data collection

2.3.1. Training and matches

The experimental period lasted two weeks and took place during the spring. Training and matches training sessions took place between 11:00 a.m. and 2:00 p.m., with a temperature range of 24 to 28 °C and relative humidity of 10 to 15%. Players performed all selected training and match sessions. Duration range for training sessions was 60 to 100 min. The duration for all matches was 90 min. All training and match sessions were performed on a similar natural turf at the university. Both selected training and match sessions consisted of four consecutive sessions, interspersed by two resting (testing) days (Fig. 1). The rating of perceived exertion (RPE), based on Borg's CR-10 scale adapted by Foster et al. [25], was collected from each participant at 30 min of the cessation of training and match sessions. Each player's rating was made individually to minimize the influence of hearing or observing the rating of their peers. On this scale, the '1' means very light activity and '10' means the maximal exertion required to determine the perception of effort from the session. The rated RPE of the player was then multiplied by the time of session/time of match in minutes to sRPE as a valid indicator of internal training load [26].

2.3.2. Heart rate variability

An Elite HRV™ smartphone application and a Polar H7 HR monitor (Polar Electro Oy; Kempele, Finland) with a sampling frequency of 1000 Hz were used by each individual for measuring heart rate variability. Raw data of R-R intervals were treated with Elite HRV™ to compute heart rate variability measures. These materials have recently been validated in a study demonstrating a large agreement with Kubios HRV 2.2 software for rMSSD calculation [27]. Recording procedures

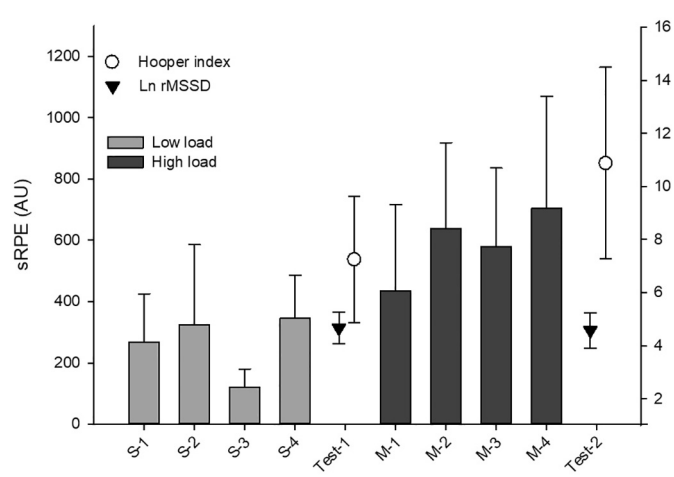

Fig. 1. Means ± SD of daily sRPE and monitoring variables during the experiment.

Note: S: Session; M: Match; sRPE: Session rating of perceived exertion; AU: arbitrary unit; Ln rMSSD: logarithm of the square root of the mean of the sum of the squares of differences between adjacent normal R-R intervals.

involved a supine HRV measurement performed by each subject with their smartphone after individual waking times (i.e., same individual time). HRV recordings lasted for 1 min, preceded by a 1.5-min stabilization period in the position [28]. All HRV recordings were completed under spontaneous breathing conditions to prevent inconsistency in Ln rMSSD results [29]. The standard deviation of normal R-R intervals (SDNN) and Ln rMSSD provided with the Elite HRV™ as useful variables for monitoring athletes' training status [2] were used for analyses.

2.3.3. Wellness measures

Subjective wellness measures were collected using a modified psychological questionnaire based on the recommendations of Hooper and Mackinnon. [17] used by previous studies [12,13,18,20,30]. The questionnaire comprises of subsets of perceived sleep quality, stress, muscle fatigue, and soreness, with each question scored on a 7-point scale (with '1' and '7' representing 'very very good' and 'very very poor' wellness ratings, respectively). Overall wellness (Hooper index) was determined by summing the four scores.

2.4. Statistical analyses

The results are presented in tables and figures as mean ± SD and/or 90% confidence intervals (CI) where specified. While actual values of heart rate variability and self-reported measures were used, data from sRPE in training and match sessions were pooled to compute their means for analyses. Within-group changes in sRPE, heart rate variability, and self-reported measures (i.e., Hooper index and its subsets) were expressed as percentage changes and as standardized differences (or standardized effect size) as Cohen's d (ES, 90% confidence limit [90% CL]) [31]. The Hopkins scale was used for ES interpretation: < 0.2: trivial; 0.2–0.6: small; 0.6–1.2: moderate; > 1.2: large [32]. Probabilities were calculated to indicate whether the true change was lower than, similar to, or higher than the smallest worthwhile change (SWC) [33]. The scale of probabilities was as follows: 25–75%: *possible*; 75–95%: *likely*; 95–99%: *very likely*; > 99%: *almost certain* [33]. The sensitivity of selected variables (i.e., Hooper index and Ln rMSSD) was assessed comparing within-test changes and between-group differences in their changes using standardized differences or effect size (ES) [33]. Pearson's correlation coefficients were used to examine the relationship between changes in mean sRPE and changes in monitoring variables. The magnitude of the correlation values was qualitatively interpreted according to Hopkins et al. [33] as follows: trivial: $r < 0.1$; small: $0.1 < r < 0.3$; moderate: $0.3 < r < 0.5$; large: $0.5 < r < 0.7$; very large: $0.7 < r < 0.9$; nearly perfect: $r > 0.9$; perfect: $r = 1$.

3. Results

Daily sRPE for low-load (i.e., training days) and high-load (i.e., match days) phases are shown in Fig. 1. Mean subjective (i.e., Hooper index) and objective (i.e., heart rate variability) measures of monitoring that were collected on days after low-load (first four days) and high-load (second four days) phases are also shown in Fig. 1. Accumulated sRPE showed an *almost certain* large increase (+148.4%, [87.3; 229.5%]), effect size (ES, 2.16 [1.49; 2.82]) from 827.1 ± 261.7 to 2026 ± 562.7 AU. A *very likely* large difference was observed in mean sRPE between low-load and high-load phases (Table 1). The Hooper index showed an *almost certainly* moderately increase after high-load phase (Table 1). Hooper index subsets also showed *likely-to-very likely* moderate increase after the high-load phase (Table 1). *Possible* trivial changes were observed in selected measures of heart rate variabilities (i.e., Ln rMSSD and SDNN) after the high-load phase (Table 1). The Hooper index showed a moderately higher sensitivity to the high-load tournament phase than did Ln rMSSD (34.7%, 90% confidence interval CI [26.9; 41.6%], effect size (ES), 0.81 [0.60; 1.03]) (Fig. 2). Changes (Δ) in mean sRPE (ΔsRPE) showed a small inverse association with ΔSDNN ($r = -0.21$, 90%CI -0.74; 0.48). Small relationship was observed between ΔsRPE and ΔLn rMSSD ($r = 0.10$, −0.56; 0.68). Small correlation was also observed between ΔsRPE and Δsoreness ($r = 0.17$, −0.51; 0.72). A small inverse relationship was found between Δstress and ΔsRPE ($r = -0.18$, −0.72; 0.50). Very large associations were observed between ΔsRPE with ΔHooper index ($r = 0.72$, 0.17; 0.93), Δfatigue ($r = 0.77$, 0.28; 0.94), and Δsleep ($r = 0.89$, 0.63; 0.98).

Table 1
Changes in training load and monitoring variables after undergoing soccer tournament.

Variable	Training phase Mean (SD) Low load	High load	% difference (90% CI)	Standardized difference (90% CI) rating	% greater/similar/lower values
Mean sRPE (AU)	287.50 (141.58)	569.38 (192.77)	108.9 (40.4; 210.9)	1.35 (0.62; 2.08) Large	99/1/0 *Very likely*
Hooper index	7.25 (2.38)	10.88 (3.60)	49.8 (33.9; 67.5)	1.05 (0.76; 1.34) Moderate	100/0/0 *Almost certain*
Sleep	2.13 (0.64)	2.88 (1.13)	29.7 (5.5; 59.5)	0.68 (0.14; 1.22) Moderate	93/6/1 *Likely*
Stress	1.25 (0.46)	1.75 (0.71)	36.4 (8.3; 71.9)	0.86 (0.22; 1.50) Moderate	95/4/1 *Likely*
Fatigue	2.13 (0.99)	3.13 (1.13)	53.0 (24.0; 88.8)	0.79 (0.40; 1.18) Moderate	99/1/0 *Very likely*
Soreness	1.75 (0.89)	3.13 (1.25)	81.9 (37.9; 140.0)	1.06 (0.57; 1.55) Moderate	99/1/0 *Very likely*
Ln rMSSD	4.67 (0.59)	4.58 (0.67)	−2.1 (−7.1; 3.1)	−0.15 (−0.50; 0.21) Trivial	4/58/40 *Possibly*
SDNN	130.07 (59.77)	138.06 (62.70)	8.2 (−7.2; 16.7)	0.15 (−0.14; 0.44) Trivial	37/60/3 *Possibly*

Note. sRPE: Session rating of perceived exertion; AU: Arbitrary unit; CI: Confidence interval; Ln rMSSD: logarithm of the square root of the mean of the sum of the squares of differences between adjacent normal R-R intervals; SDNN: Standard deviation of normal R-R intervals; SD: Standard deviation.

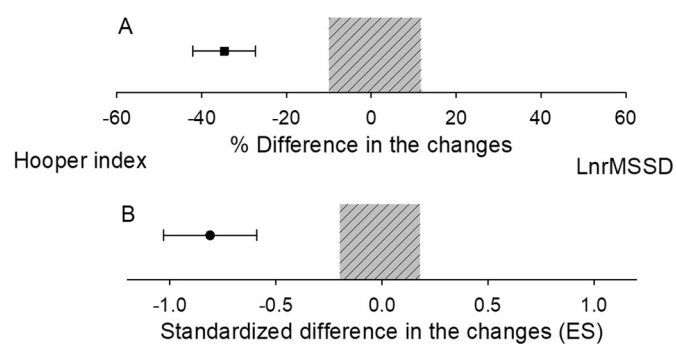

Fig. 2. Differences in the changes (90% confidence intervals) are expressed as (A) percentage or (B) standardized differences. Shaded areas represent the range of trivial change or difference (see Methods).
Note. Ln rMSSD: logarithm of the square root of the mean of the sum of the squares of differences between adjacent normal R-R intervals; ES; Effect size.

4. Discussion

The aims of this study were a) to examine within-group changes in wellness and heart rate variability measures and b) to compare their sensitivity to a congested match schedule in collegiate soccer players. The main finding of the present study was that, while wellness measures moderately changed due to a large increase in mean sRPE, heart

rate variability indices only showed a trivial change over four consecutive days of monitoring. Relationships between changes in the Hooper index and some of its subsets (ΔHooper index, Δsleep, and Δfatigue) with changes in mean sRPE (ΔsRPE) were very large (range $r = 0.72$; 0.89). However, small associations were observed between changes in heart rate variability (ΔLn rMSSD, and ΔSDNN) and ΔsRPE (range $r = -0.21$; 0.10).

Mean sRPE showed a large increase in congested collegiate match schedule compared with the low-load training phase prior to the tournament in the present study (Fig. 1). These results are in agreement with previous findings of higher training loads during a fixture match schedule in soccer [13]. While professional soccer players usually undergo an intensified preparatory phase before starting their yearly competitions [16,18,34], collegiate-level soccer players in some countries (e.g., Iran) participate in easy-to-moderate technical/tactical training sessions aimed to achieve a greater tactical coordination before commencing tournaments.

Our results showed that the Hooper index moderately increases over a congested match schedule in collegiate soccer players. This finding confirms the results of previous studies which have shown a detrimental effect of increased training load on Hooper index values [13,18,20]. However, the change magnitude of Hooper index values observed here is larger than the value reported by Clemente et al. [13] (ES, 1.05 versus 0.03). This larger change might be related to the higher number of weekly matches played in the present study compared to the previous study (n = 4 vs. 2).

The moderate sleep quality impairment observed in the present study is in line with previous studies which have reported the detrimental effect of a congested match schedule [13] and high-load training phases [18] on this measure. The *very large* association between Δsleep with ΔsRPE ($r = 0.89$) found in this study is in agreement with the findings of the study by Molla et al. [20], which showed significant relationships between this measure and training load. These results are not, however, consistent with the study of Thorpe et al. [8], in which a significant relationship between sleep quality and training load was not observed. It seems that different training load variables (internal or external training load) are, at least in part, responsible for this contradiction. Indeed, in the study conducted by Molla et al. [20] and in our study, sRPE has been used as an estimate of internal training load; Thorpe et al. [8], conversely, monitored total high-intensity running distance as an external training load variable. However, since sleep quality might be influenced by many factors [35], including match results [36], further investigations are needed to clarify the sensitivity of self-reported sleep quality to training loads.

An increase in stress, induced by a moderately congested match schedule, was observed in the present study (ES: 0.86). This confirms previous reports of stress escalation due to high versus low training loads [13,18–20]. The larger magnitude of stress escalation observed in the present study compared with the results of Clemente et al. [13] (moderate versus trivial) might also be related to the number of matches (n = 2 versus 4) performed. Therefore, our results, consistent with previous findings, suggest that observing escalated stress can be expected in soccer players when they undergo high-load training phases. The escalated stress is likely related to increased anxiety in higher versus lower training loads [37].

Fatigue also showed a moderate impairment during the congested schedule in the present study, which supports previous findings of increased fatigue due to congested match schedules [13], high-load training phases [16,18,19], and daily training load fluctuations [8,20]. The magnitude of fatigue change observed in the present study (0.79) is, however, higher than values reported by Flatt et al. [16] (ES: 0.45), and Clemente et al. [13] (ES: 0.06) but lower than changes found by Flatt et al. [19] (ES: −1.41 and 1.98) and Fessi et al. [18] (ES > 2.0). The larger impairment in fatigue found in this study compared to the results of Clemente et al. [13] is likely due to the higher number of matches during congested match schedule in this study (n = 4 versus 2). The lower magnitude of fatigue change observed here compared with that observed by Flatt et al. [19] and Fessi et al. [18] might be pertinent to different sports of participants (sprint-swimmers versus soccer players) and between test periods (eight weeks versus five days), respectively.

Soreness also showed a moderate impairment (ES: 1.06) after undergoing the high-load phase in this study. This result is consistent with the results of previous studies which have shown an impairment in soreness when training load was increased [13,18,19,21] but not with other studies which reported unchanged soreness scores [16]. The magnitude of soreness impairment observed in the present study is higher than values reported by Clemente et al. [13] but almost similar to the results reported by Lundberg and Weckström [21]. This might be due to the more highly-congested match schedule in the study of Lundberg and Weckström [21] than in the study conducted by Clemente et al. [13] (3 versus 2 weekly matches). Subsequently, it seems that as the number of matches during a congested match schedules increases, the impairment in fatigue also increases.

Our results also showed that selected heart rate variability measures result in only a *possibly* trivial change to the congested schedule. The results of the present study contrast results of previous studies which have shown significant changes in Ln rMSSD in high-load versus low-load training phases [16,19,22] and another study that found significant relationships between Ln rMSSD and fluctuations in training loads [8]. The reason for this discrepancy is likely differences in the methodological approaches used. Previous studies used rolling average or coefficient of variations of Ln rMSSD (i.e., Ln $rMSSD_{mean}$ and Ln $rMSSD_{cv}$(derived from multiple time points (i.e., 3 to 7 time points) [38–41], whereas only one time point was used for data collection in the present study. This result confirms previous recommendations for computing a rolling average with at least three weekly time points for monitoring training status when measuring heart rate variability [41]. However, the aim of the present study was to measure the sensitivity of Ln rMSSD compared with self-reported measures for a congested match schedule at only one time point. Indeed, during a congested match schedule, coaches are usually interested in finding players' fatigue conditions after three to four consecutive matches to make decisions for the lineup of upcoming matches. SDNN was also used as a measure of heart rate variability in the present study. While it has been confirmed that this index can also be used for monitoring training status of athletes [2], the same trend (trivial *change*) was observed in this index. However, since, to our knowledge, no study has used changes of this measure due to congested match schedule, no valid comparison can be made.

Finally, and more importantly, the Hooper index showed a moderately larger sensitivity to the high-load training phase than Ln rMSSD did in the present study (Fig. 2). A very large association between ΔsRPE and ΔHooper index ($r = 0.72$) compared with the small relationship between ΔsRPE and ΔLn rMSSD ($r = 0.10$) again suggest a higher sensitivity of subjective self-reported measures when monitoring soccer players' fatigue status during congested match schedules. These results confirm the finding of recent review which lhas shown the superiority of subjective measures over objective indices [42]. Among subsets of the Hooper index, sleep and fatigue also showed *very large* prediction abilities in increases of training load. Therefore, the results of this study suggest that it is more effective to use subjective self-reported measures of wellness instead of heart rate variability indices when "one time point" is the only option available.

This study had some limitations. External load was not measured in the present study. The size of the sample was small, the observation period was quite brief and the competitive level was not professional. Therefore, our conclusions are also limited. However, the results of this study are promising regarding the usefulness of wellness questionnaires to assess important factors associated with training and match load. This is in line with the importance of monitoring wellness status in both professional and non-professional teams by using easy-to-use and cost-

effective methods. Future studies investigating time-course of recovery after matches using subjective and objective measures can also help to determine which measures are more useful when monitoring players' training-induced fatigue over longer periods.

5. Conclusions

In conclusion, based on the findings from this study, it is suggested to use subjective wellness measures instead of heart rate variability indices when "one time point" is efficient option available to monitor soccer players' training status during congested match schedules. The results found more sensitiveness of this measure to variations of load than HRV. Hooper index and its subsets including quality of sleep, stress, fatigue and soreness are very time-efficient and easy-to-use measures to monitor athletes' training status during high-load phases. Therefore, while practitioners can use both HRV and wellness measures to monitor soccer players' fatigue, they are encouraged to use the Hooper index as the most sensitive measure when "one-time point" is the only option.

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Declarations of interest

None.

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